Normalised almost-Besse contact forms: boundary behaviour and C1-closeness of the Reeb field

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Editorial note. The counted statement is the author's labelled lemma 'l:nearly_Besse_normalized' (source lines 1289-1310); the proof environment begins on the next line after \end{lem} apart from one blank line, so the pairing is by adjacency.

Editorial note. No context span is retained: the statement and proof are self-contained in the pinned author file, and the notation they use is either defined inside them or carried by the retained author macro block.

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Editorial scope. One original result of the pinned publisher article, counted once: the normalisation lemma for almost-Besse contact forms (author label l:nearly_Besse_normalized).

Allowed standard prerequisites (editorial).

- Almost-Besse contact form. A contact form is almost-Besse when the Reeb field is nowhere tangent to the contact structure and the linearised return map along each closed Reeb orbit has all eigenvalues real.. It is the hypothesis under which the four conclusions of the statement are derived.
- Collar coordinates on the boundary. A collar neighbourhood A_C x S^1 of a component C x S^1 of the boundary is identified with [0,rho) x S^1 x S^1, and the solid cylinder Y_1 with B_1 x S^1.. They are introduced at the start of the proof at source line 1313 to write the gluing map explicitly.
- The map tilde-iota and the normalised form. The map tilde-iota(r,s,t) = (r e^{2 pi i (q_0 s - alpha_1' t)}, p_0 s + alpha_1 t) identifies the collar with the cylinder, and tilde-lambda_0 is the displayed normalised contact form.. They are used at source lines 1313-1327 to compute the pull-back of nu and the vanishing of omega.

Lemma 0.1. *i. The pull-back of ν via the inclusion $\partial\Sigma \hookrightarrow \Sigma$ satisfies*

$$\nu|_{\partial\Sigma} = \nu_0|_{\partial\Sigma}.$$

ii. The 2-form ω vanishes at all points of $\partial\Sigma$, i.e.

$$\omega_z = 0, \qquad \forall z \in \partial\Sigma.$$

iii. The Reeb vector field $R_{\tilde\lambda}$, a priori only defined on the interior of $\Sigma \times S^1$, admits a smooth extension to $\Sigma \times S^1$ that is tangent to the boundary $\partial\Sigma \times S^1$.

iv. For all $\varepsilon > 0$ there exists $\delta > 0$ such that, if the above contact form λ further satisfies $\|R_\lambda - R_{\lambda_0}\|_{C^2} < \delta$, then $\|R_{\tilde\lambda} - \partial_t\|_{C^1} < \varepsilon$. Here, the norms are associated with arbitrary fixed Riemannian metrics.

Proof. Let us consider a collar neighborhood $A_C \times S^1$ of a connected component $C \times S^1$ of $\partial\Sigma \times S^1$, and the solid cylinder neighborhood $Y_1 \subset Y$ of γ_1. With the identifications $A_C \equiv [0,\rho) \times S^1$, $Y_1 \equiv B_1 \times S^1$, and $\gamma_1 \equiv \{0\} \times S^1$, as in the previous subsection, the restriction of the map $\tilde\iota$ can be written as

$$\tilde\iota|_{A_C\times S^1} : A_C \times S^1 \to Y_1 \equiv B_1 \times S^1, \qquad \tilde\iota(r,s,t) = \big(re^{2\pi(q_0 s - \alpha_1' t)i}, p_0 s + \alpha_1 t\big),$$

and $\tilde\lambda_0|_{A_C\times S^1}$ can be written as

$$\tilde\lambda_0|_{A_C\times S^1} := dt + \Big(\frac{p_0}{\alpha_1} + \frac{2\pi}{\alpha_1}\big(q_0 + (\alpha_1'/\alpha_1)p_0\big)r^2\Big)ds. \tag{1}$$

The computations

$$\tilde\iota_*\partial_t|_{C\times S^1} = R_{\lambda_0} = R_\lambda, \qquad \tilde\iota_*\partial_s|_{C\times S^1} = p_0\partial_s = \frac{p_0}{\alpha_1}\, R_{\lambda_0} = \frac{p_0}{\alpha_1}\, R_\lambda,$$

imply

$$\tilde\lambda(\partial_t)|_{C\times S^1} = \tilde\lambda_0(\partial_t)|_{C\times S^1} \equiv 1, \qquad \tilde\lambda(\partial_s)|_{C\times S^1} = \tilde\lambda_0(\partial_s)|_{C\times S^1} \equiv \frac{p_0}{\alpha_1},$$

and

$$\begin{aligned} (\partial_t \lrcorner\, d\tilde\lambda)_z &= \tilde\iota^*(R_{\lambda_0} \lrcorner\, d\lambda)_z = \tilde\iota^*(R_\lambda \lrcorner\, d\lambda)_z \equiv 0, \\ (\partial_s \lrcorner\, d\tilde\lambda)_z &= \frac{p_0}{\alpha_1}\,\tilde\iota^*(R_{\lambda_0} \lrcorner\, d\lambda)_z = \frac{p_0}{\alpha_1}\,\tilde\iota^*(R_\lambda \lrcorner\, d\lambda)_z \equiv 0, \end{aligned} \qquad \forall z \in C \times S^1.$$

These identities, together with $\nu_0 = \tilde\lambda_0|_{\Sigma\times\{0\}}$ and $\nu = \tilde\lambda|_{\Sigma\times\{0\}}$, readily imply points (i) and (ii)

As for point (iii), let us write $R_\lambda|_{Y_1}$ in coordinates $(x+iy, s) = (re^{i\theta}, s)$ as

$$\begin{aligned} R_\lambda|_{Y_1} &= F_1\partial_x + F_2\partial_y + \partial_s \\ &= (F_1\cos(\theta) + F_2\sin(\theta))\,\partial_r + \frac{1}{r}\,(F_2\cos(\theta) - F_1\sin(\theta))\partial_\theta + F_3\,\partial_s, \end{aligned}$$

for some smooth functions $F_j : Y_1 \to \mathbb{R}$. Since R_λ and R_{λ_0} coincide along the closed Reeb orbit γ_1, we have

$$F_1(0) = F_2(0) = 0, \qquad F_3(0) = \alpha_1.$$

Since

$$\tilde{\iota}_*\partial_r = \partial_r, \qquad \tilde{\iota}_*\partial_s = 2\pi q_0\,\partial_\theta + p_0\,\partial_s, \qquad \iota_*\partial_t = -2\pi\alpha_1'\,\partial_\theta + \alpha_1\,\partial_s,$$

the Reeb vector fields $R_{\tilde{\lambda}}$ on the interior $\mathrm{int}(A_C)\times S^1$ is given by

$$R_{\tilde{\lambda}} = f_1\,\partial_r + f_2\,\partial_s + f_3\,\partial_t,$$

where, if we set

$$\theta(s,t) := q_0 s - \alpha_1' t, \qquad G_j(r,s,t) := \frac{F_j\circ\iota(r,s,t)}{r}, \quad j=1,2,$$

the functions $f_j : \mathrm{int}(A_C)\times S^1 \to \mathbb{R}$ are given by

$$f_1 = \cos(\theta)F_1\circ\tilde{\iota} + \sin(\theta)F_2\circ\tilde{\iota}, \tag{2}$$

$$f_2 = \big(2\pi\big(q_0 + (\alpha_1'/\alpha_1)p_0\big)\big)^{-1}\big(-\sin(\theta)G_1 + \cos(\theta)G_2 + 2\pi(\alpha_1'/\alpha_1)F_3\circ\tilde{\iota}\big), \tag{3}$$

$$f_3 = \frac{1}{\alpha_1}\big(F_3\circ\tilde{\iota} - p_0\,f_2\big). \tag{4}$$

Since $F_1\circ\tilde{\iota}|_{C\times S^1} = F_2\circ\tilde{\iota}|_{C\times S^1}\equiv 0$, the functions G_1 and G_2 extend smoothly to the whole $A_C\times S^1$, and so do the functions f_1, f_2, f_3. Moreover, $f_1|_{C\times S^1}\equiv 0$. This proves that $R_{\tilde{\lambda}}$ extends smoothly to a vector field on $\Sigma\times S^1$ that is tangent to the boundary $\partial\Sigma\times S^1$.

Finally, assume that R_λ is C^2-close to R_{λ_0}. Away from any fixed neighborhood of $\partial\Sigma\times S^1$, $R_{\tilde{\lambda}}$ is C^2-close to $R_{\tilde{\lambda}_0} = \partial_t$. On Y_1, since $R_\lambda = -2\pi\alpha_1'\partial_\theta + \alpha_1\partial_s$, the functions

$$F_1 - 2\pi\alpha_1' r\sin(\theta), \qquad F_2 + 2\pi\alpha_1' r\cos(\theta), \qquad F_3 - \alpha_1$$

are C^2-small and vanish at $\partial\Sigma\times S^1$. Therefore, the function

$$-\sin(\theta)G_1 + \cos(\theta)G_2 + 2\pi(\alpha_1'/\alpha_1)F_3\circ\tilde{\iota}$$

is C^1-small. Equation (2) implies that f_1 is C^2-small, and Equations (3) and (4) imply that f_2 and $f_3 - 1$ are C^1-small. Overall, we conclude that $R_{\tilde{\lambda}}$ is C^1-close to $R_{\tilde{\lambda}_0} = \partial_t$. □

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